

Air Quality Analytics and Insight Report

Project 01 — Data Analysis Foundation | MenuManager ML Internship

Intern	Darshan Suresh S
Dataset	UCI Air Quality Data Set (9,358 hourly records)
Tools	Python, Pandas, NumPy, Matplotlib, Jupyter
Report scope	Problem, method, findings, limitations, conclusion

1. Problem Statement

Urban air-quality sensor data is often noisy, incomplete, and difficult to interpret directly. This project transforms raw hourly multisensor readings from an Italian city (March 2004 - February 2005) into a clean analytical dataset, and produces evidence-based insights about pollution patterns, environmental conditions, and data quality. The dataset contains 9,358 hourly records with missing values encoded as the sentinel value -200, sourced from the UCI Machine Learning Repository.

2. Method

2.1 Data Loading and Audit

The raw file is semicolon-delimited with European decimal-comma formatting and includes two trailing empty columns plus a block of fully empty rows at the end of the export. After removing these artifacts, 9,357 real hourly records remain, with no exact duplicate rows.

2.2 Datetime Features

Date (DD/MM/YYYY) and Time (HH.MM.SS) were combined into a single datetime column, from which hour, day, month, weekday name, and an is_weekend flag were derived to support time-based aggregation.

2.3 Missing-Value Treatment

The -200 sentinel was replaced with NaN across all numeric columns. Treatment was then chosen per column based on missingness level and pattern:

Column group	Missing %	Treatment	Justification
NMHC_GT	~90%	Dropped entirely	Too sparse to impute credibly
CO_GT, NOx_GT, NO2_GT	~17-18%	Time interpolation (6h limit) + hour-of-day median fallback	Missingness occurs in contiguous outage blocks, not at random
Remaining sensors/weather	~4%	Same as above	Short, scattered device-reset gaps

This avoids the common error of filling a fast-changing, cyclical time series with a single global mean, which would flatten real diurnal and seasonal pollution patterns.

2.4 Statistics and Custom Metrics

NumPy was used to compute mean, median, standard deviation, percentiles (10th/25th/75th/90th), and IQR-based outlier thresholds for every cleaned column. Two custom metrics were derived: a composite **Pollution Index** (equal-weighted, min-max normalized average of CO, NOx, and NO2 ground-truth readings) and a **Sensor Reliability Ratio** (metal-oxide CO sensor reading divided by the certified analyzer reading), used to flag potential sensor drift.

2.5 Aggregation

Pandas groupby, pivot_table, resample, and corr operations were used to compare pollution levels by hour, month, and weekday vs. weekend, and to quantify relationships between pollutants and weather variables.

3. Findings

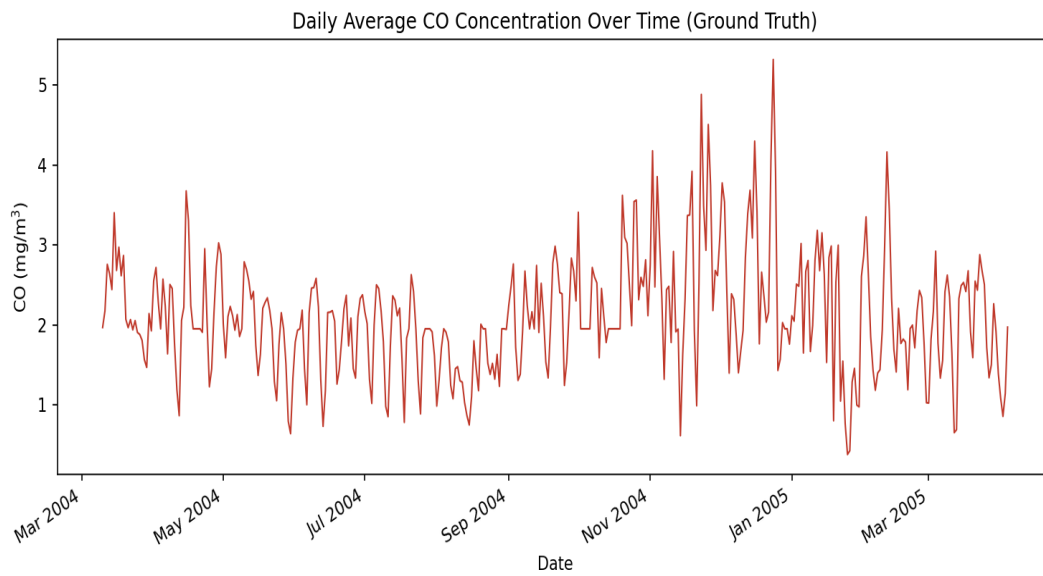


Figure 1. Daily average CO shows clear seasonal structure across the full monitoring period.

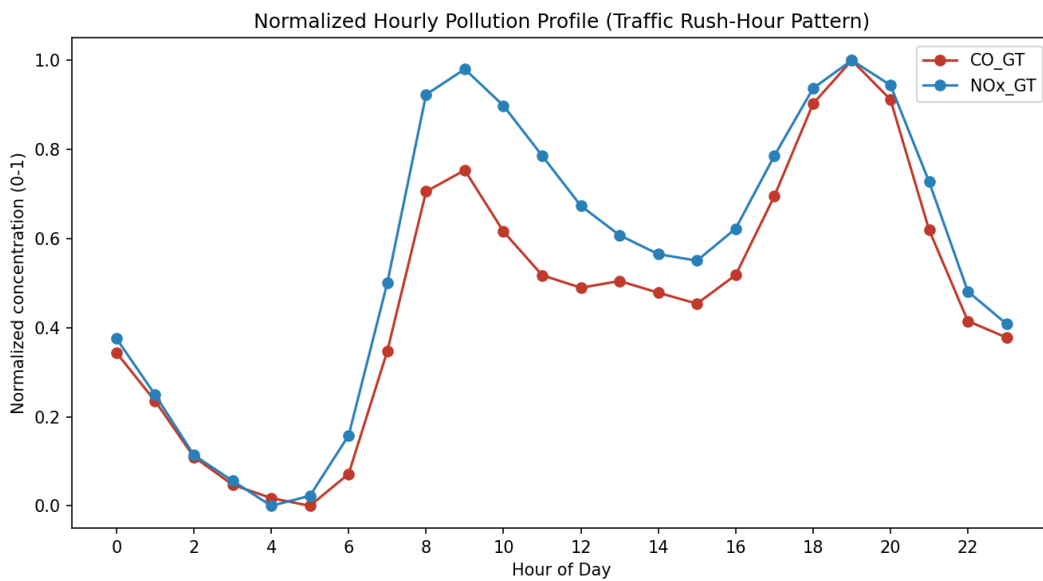


Figure 2. Normalized hourly profile reveals a traffic-driven morning/evening double peak for CO and NOx.

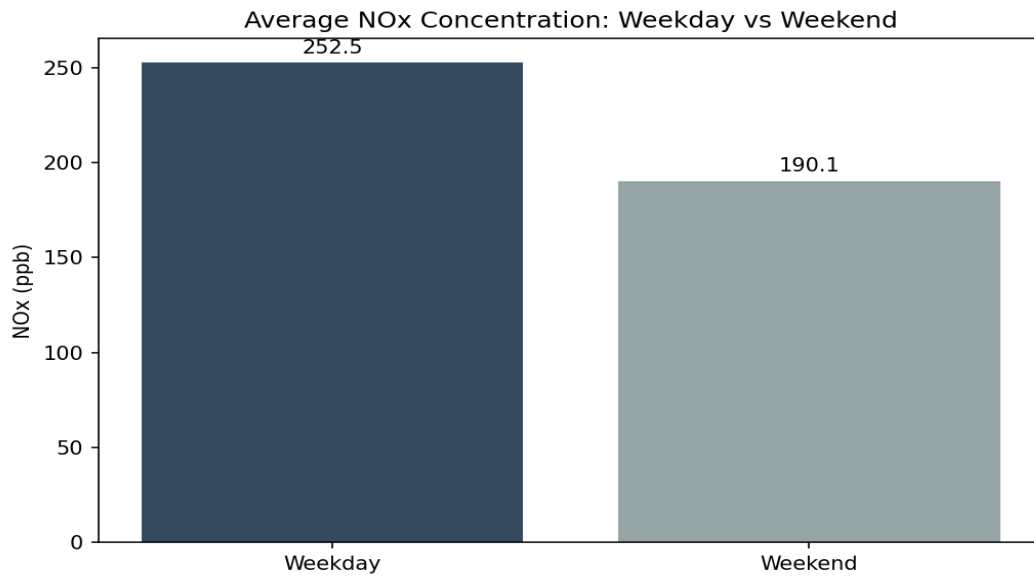


Figure 3. Weekday NOx is substantially higher than weekend NOx, consistent with commuter traffic.

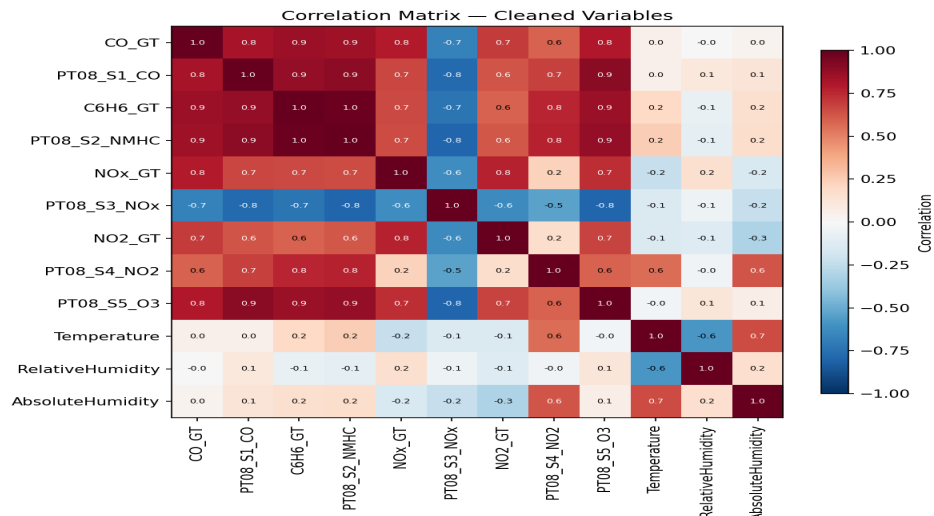


Figure 4. Correlation matrix across cleaned pollutant and weather variables.

Key Observations

- 1 CO, NOx, and benzene all show a clear double peak on weekdays around 08:00 and 19:00-20:00, matching morning and evening commuter traffic rather than random background noise.
- 2 Weekday NOx (~252 ppb average) is about 33% higher than weekend NOx (~190 ppb), reinforcing that traffic is the dominant local NOx source.
- 3 Winter months (Nov-Feb) show consistently higher CO than summer months (Jun-Aug), likely from heating emissions plus more stagnant, cooler air trapping pollutants near ground level.
- 4 The metal-oxide CO sensor (PT08.S1) correlates strongly (r approx. 0.8-0.9) with the certified analyzer, confirming it is a usable low-cost proxy for CO, though the relationship shows non-linearity at high concentrations, consistent with typical sensor saturation behavior.

- 5 Relative humidity and temperature are strongly negatively correlated (r approx. -0.6), the expected physical relationship, providing a useful sanity check on sensor data quality.

4. Limitations

- 1 NMHC_GT was approximately 90% missing and had to be dropped entirely rather than imputed; no conclusion about non-methane hydrocarbons specifically can be drawn from this dataset.
- 2 Missing values in the remaining columns occur in contiguous outage blocks rather than completely at random. Interpolation and hour-of-day medians are reasonable compromises, but necessarily smooth over whatever real short-term variation happened during those outages.
- 3 The dataset covers a single monitoring site over one year; findings describe this location's traffic-driven pollution pattern and should not be generalized to other cities without further validation.

5. Conclusion

The raw hourly sensor export was audited, cleaned with a documented, column-specific missing-value strategy, enriched with calendar features and two custom metrics, and explored through nine visualizations. The dominant pattern across the dataset is traffic-driven pollution: clear rush-hour peaks, weekday-vs-weekend gaps, and winter-vs-summer differences all point to vehicle emissions and seasonal heating as the primary drivers at this monitoring site, with the metal-oxide sensors behaving as usable, if imperfect, low-cost proxies for the certified reference measurements. The cleaned dataset (`cleaned_air_quality.csv`) and summary statistics table (`summary_statistics.csv`) are included alongside this report for reproducibility.

Dataset citation: De Vito, S. (2016). Air Quality [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C59K5F>